



Oil volatility and sovereign risk of BRICS

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ABSTRACT

We study the dependence between oil implied volatility shocks and BRICS sovereign risk from July 2009 to March 2017. First, we examine spillovers in value at risk using multivariate regression quantiles and reveal that oil volatility represents a common risk for oil-exporting and oil-importing BRICS countries. We also employ a quantile impulse-response function and reveal the presence of an asymmetry in the mechanisms of shock transmissions between oil exporters (Russia and Brazil) and oil importers (China and India): the former are more sensitive to positive oil shocks, whereas the latter are more sensitive to negative oil shocks. Second, we measure the directional predictability in the quantiles using the bivariate cross-quantilegram approach and show that in most cases, a low- (high-) volatility of oil market predicts low (high) sovereign risk at various quantiles and lags. Policy implications are discussed.

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1. Introduction

The globalization of markets has contributed to the increase in the demand for sovereign credit risk instruments, such as Credit Default Swap (CDS) which is a bilateral contract between a buyer and a seller where the buyer pays a periodic premium for insurance against a debt default. Typically, it applies to government and corporate bonds and is used by banks and hedge funds to hedge credit risk exposures, particularly in emerging economies that have low credit ratings, have not achieved a higher level of diversification, and still depend on energy exportation (UNDP, 2011). These features are also relevant to (some of) the large emerging markets that attract foreign institutional investors and portfolio managers whose primary focus is maximizing risk–return tradeoffs through diversification. A change in the sovereign risk quality can pose challenges for businesses and governmental bodies and, in particular, affect the re-weighting of international portfolios. Consequently, investors and risk managers would benefit from understanding the factors that affect sovereign credit risk.

While it is well documented that sovereign risk is affected by several key factors, such as macroeconomic instabilities, institutional quality of the country, external debt, foreign reserves, inflation, and economic growth, it can also be significantly affected by the volatility of oil prices, especially in large emerging economies. Bhar and Nikolova (2009) argue that emerging economies are more vulnerable to oil price volatility than developed countries because they grow more rapidly and have more energy-intensive production structures.¹ In some emerging economies that are dependent on oil exportation, such as Russia, oil price volatility leads to macroeconomic instabilities as well as to a volatility in export earnings, foreign exchange reserves, current account balances, economic activities, and eventually to difficulties in meeting debt obligations. Hilscher and Nosbusch (2010) argue that a country with historically higher macroeconomic volatility is more prone to default. Notably, an increase in the cost of government borrowing is reflected in higher CDS spreads and potentially leads to a sovereign debt default (Hilscher and Nosbusch, 2010). Hence, the effect of oil price volatility on sovereign credit risk can represent an important source of instabilities in macroeconomic fluctuations that both policy-makers and practitioners may want

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¹ This is in line with Ang and Longstaff (2013) who argue that the systematic risk component of the sovereign risk spread is affected by global risk factors that include energy price fluctuations.

to respond to and consider (Liu et al., 2016). In particular, the recent sharp decrease in crude oil prices in late 2014 coupled with high oil price volatility has increased the focus on the oil-sovereign credit risk link. In 2015 and 2016, several press articles were released that demonstrated how falling oil prices and increasing oil volatility were closely linked to the quality of sovereign credit risk, especially in heavily-oil-dependent economies, such as Russia, that are strained in funding their public budget and paying off their external dollar-denominated debt. Although several studies consider the predictive power of oil prices – which are part of commodities prices – in modeling the sovereign yield spread in emerging economies (Longstaff et al., 2011; Amstad et al., 2016; Aizenman et al., 2016), the effect of oil price volatility² on sovereign CDS spreads remains extremely understudied. Against this background, the aim of this study is to examine the dependence between oil price volatility and BRICS³ sovereign risk using a rich and dynamic approach that is capable of capturing upside and downside risk spillovers from oil volatility to BRICS CDS spreads and measuring within a cross-quantilogram (CQ) bivariate model the predictability of BRICS sovereign risk based on oil volatility in the quantiles. Our choice for large emerging countries, i.e. BRICS, stems from the heterogeneity of this group, inclusive of both large oil-importing countries (China and India) and large oil-exporting countries (such as Russia), and therefore the possibility to uncover through our empirical analyses any significant differences in the impact of oil price volatility on the sovereign risk between oil-exporting and oil-importing countries.⁴

Our study is related to the limited empirical literature on the oil volatility-sovereign risk link (Sharma and Thuraissamy, 2013; Liu et al., 2016; Bouri et al., 2017a; Lee et al., 2017). Importantly, it addresses several shortcomings and explores new empirical grounds. Sharma and Thuraissamy (2013) highlight the role of oil price volatility in predicting sovereign CDS returns in six Asian countries (Indonesia, Japan, Malaysia, the Philippines, South Korea and Vietnam). However, they overlook the effect of oil uncertainty on BRICS sovereign risk and more importantly do not account for the codependence between sovereign risk and oil implied volatility in the different quantiles. Liu et al. (2016) focus on the statistical properties of country risk rating in oil-exporting countries and show that oil price volatility can accentuate the volatility of country risk ratings. The authors cover only oil-exporting countries within their analysis and use relatively low-frequency (monthly) data on country risk from the International Country Risk Guide. In addition, their focus is limited to the mean-based relationships between the examined variables. Lee et al. (2017) show that country risk in net oil-importing (Germany, France, Italy, Japan and US) and net oil-exporting countries (Canada, UK) is affected by oil price shocks. However, the authors overlook, among other things, oil importers and oil exporters from BRICS as well as the effect of oil (implied) volatility shocks across the quantiles. As indicated by several studies (e.g., Ville and Wicken, 2012; Holden, 2013), advanced economies have achieved higher level of diversification, have an AAA credit rating, and have escaped the so-called “resource curse”, which suggests that the examination of the oil volatility-sovereign risk in emerging economies is more relevant. Bouri et al. (2017a) use a GARCH-based measure of oil volatility and concentrate on the average causal relationship between the volatilities of energy commodities and CDS spreads without taking into account

the difference in the dependence between high and low quantiles. It is documented that the return and volatility distribution of assets is most often heterogeneous, which suggests that it is appropriate to use a quantile-based approach to uncover differences in the response of sovereign risk to oil volatility shocks across the different quantiles. We follow the recently proposed methodology of White et al. (2015) and address this empirical gap. White et al. (2015) proposed an interesting model called VAR for VaR – that is, vector autoregressive (VAR) for value at risk (VaR) – that can assess the codependence between BRICS sovereign CDS spreads and oil implied volatility. Their proposed model contains multivariate regression quantiles that overcome the need for strong distributional assumptions given that it directly models the (stationary) series. Compared to univariate quantile models, it uncovers richer dynamics, and it allows for estimating a VAR process on the quantiles of the stationary series – in our case, CDS spreads and oil implied volatility – rather on the mean. By applying the VAR for VaR, we can also focus on the upside and downside risk spillovers from oil volatility to BRICS CDS spreads and calculate pseudo quantile impulse-response functions (QIRF). Finally, we apply the CQs of Han et al. (2016) to capture the directional quantile predictability of BRICS CDS spreads based on oil implied volatility.⁵ The adopted empirical methods enrich our analysis by implicitly accounting for the impact of the large volatility experienced by the oil market during the European debt crisis of 2010–13 and oil slump of mid-2014 on BRICS sovereign risk.

In assessing the vulnerability – or, alternatively, the resistance – of BRICS sovereign risk to oil uncertainty, we seek to uncover BRICS's economic heterogeneity vis-à-vis oil market volatility. This is relevant to the structure of our sample, which covers two major importers and consumers of crude oil (China and India) and one of the leading oil producers and exporters (Russia).⁶ Intuitively, the effect of volatility price shocks on sovereign risk should be larger in oil-exporting countries than in oil-importing countries.

The study is organized as follows. The next section provides an overview of the related literature, Section 3 presents the econometric techniques that we employ, and Section 4 describes the data and presents the empirical results. Finally, Sections 5 concludes.

2. Literature review

In this section, we explore two main strands of research that are closely related to the focus of our study. The first is price discovery in the sovereign CDS market. The second is the effect of oil implied volatility on financial variables.

Along with globalization and the rising importance of the sovereign CDS in mitigating credit risk, numerous studies have attempted to understand the determinants of sovereign CDS spreads in large emerging countries that attract important international capital. Some studies focus on the role played by commodities, including crude oil. Longstaff et al. (2011) show that oil and commodity price changes are among the main contributors to the sovereign CDS spreads of 26 developed and emerging-market countries. Aizenman et al. (2016) concentrate on the sovereign risk of emerging markets and show that the commodity term of trade volatility is positively associated with sovereign CDS spreads. Few other studies include the impact of oil prices while modeling sovereign spreads. Duffie et al. (2003) show that Russia's credit risk responds to political events and is negatively correlated with the country's foreign currency reserves and oil prices. Alexandre and de

² According to Bouri et al. (2017a), it would be more interesting to relate the sovereign risk with the commodity (oil) price volatility than with the oil price level given that that the former is more explicitly associated with the volatility of the country's macroeconomic fundamentals. For example an increase in the volatility of macroeconomic fundamentals can increase the probability of sovereign debt default.

³ BRICS stands for five major emerging economies: Brazil, Russia, India, China and South Africa. The first four countries were grouped as “BRIC”, before the induction of South Africa in 2010. BRICS include >40% of world population and around 30% of world economic output.

⁴ We chose to overlook developed economies – although they include major oil-exporting countries – and focus only on emerging economies because emerging economies have lower credit ratings, have not achieved a higher level of diversification, and still depend on energy exportation.

⁵ Although prior studies considered the dependence between two variables based on copula and GARCH, the dependence between sovereign risk and oil implied volatility in BRICS remains understudied – especially from the perspective of multivariate quantile regression and directional predictability.

⁶ According to the latest figures from IndexMundi, China (India) imported and consumed, respectively, 5664 (3272) and 10,480 (3660) million barrels of crude oil per day in 2013. Notably, China surpassed the US and become the largest oil importer in 2015, with a record 6.7 million barrels imported from overseas. Russia is the second largest producer and exporter of crude oil after Saudi Arabia.

Benoist (2010) examine the impact of oil price returns and volatility on bond risk premiums in seventeen emerging countries from 1998 to 2008. The authors stress the important role played by oil price return and volatility in determining sovereign credit risk, especially in oil-exporting countries (i.e., Russia and Brazil). Fonseca et al. (2016) stress the importance of volatility and jumps extracted from the oil futures in explaining CDS spread changes in the energy sector. Sharma and Thuraissamy (2013) focus on six Asian countries (Indonesia, Japan, Malaysia, the Philippines, South Korea and Vietnam) and find that oil price uncertainty can be used as a predictor for CDS returns in several cases. Instead, we concentrate on the effect of oil volatility, as measured by the CBOE oil implied volatility index (OVX), on the volatility of CDS spreads. Hooper (2015) examines the association between oil and gas reserves and sovereign spreads in 10 emerging oil-exporting countries from 1994 to 2014. The author shows that oil reserves have an effect on sovereign spreads but that this effect also depends on the institutional quality of the country, which is determined by its corruption, political stability, and democracy. Liu et al. (2016) incorporate the variable of oil price volatility into the theoretical framework of Cruces (2006) and find that the statistical properties of country risk among oil-exporting countries remain steady despite oil price volatility. Remarkably, the authors show that oil price volatility can accentuate the volatility of country risk ratings. Lee et al. (2017) use a structural VAR approach to analyze monthly data and show that country risk in net oil-importing (Germany, France, Italy, Japan and US) and net oil-exporting countries (Canada, UK) is affected by oil price shocks. We instead focus on sovereign risk, as measured by CDS spreads, using a daily dataset. We also account for the impact of oil volatility on the different quantiles of sovereign risk. In an interesting paper, Bouri et al. (2017a) explicitly examine the volatility transmission from commodity and energy indices to sovereign CDS spreads in emerging and frontier economies. Based on a causality-in-variance test, the authors reveal that the volatility transmission from commodities to CDS spreads is not clear cut as it does not depend solely on commodity and energy dependence. However, Bouri et al. (2017a) do not differentiate between upper and lower tail dependence in the dynamics of the causality in variance, but they instead focus only on a mean-based measure of volatility linkages. We contribute to the finance literature by uncovering the tail interdependence between oil implied volatility and sovereign CDS spreads of BRICS countries under different market conditions (i.e., in lower, median, and upper quantiles), considering for the response of CDS sovereign risk to oil volatility shocks using quantile IRFs (White et al., 2015).

Given our reliance on implied volatility, which is a forward-looking measure of volatility, we also focus on the directional predictability from the perspective of investors' expectations of oil volatility within the predictive approach of Han et al. (2016), which allows for measuring directional predictability in quantiles from oil implied volatility to BRICS sovereign risk.

In contrast to prior studies, which usually use a backward-looking proxy of volatility such as absolute returns, squared returns, or GARCH-based estimates, we employ the market's expectations of 30-day volatility based on option data. In addition to historical volatility information, implied volatility reflects investors' expectations of future market conditions. Given that market participants are more concerned with future volatility measures than historical volatility measures, the inclusion of implied volatility enriches our empirical analysis. Recently, the impact of the CBOE OVX on some other financial variables has been the subject of a few studies. Liu et al. (2013) focus on the short- and long-run relationships between the implied volatility of oil and that of US equities, the euro/dollar exchange rate, and gold prices. Maghyereh et al. (2016) use implied volatility indices within the oil volatility-stock nexus. Bouri et al. (2017b) examine the effect of the implied volatility of oil on that of gold and Indian equities. As shown, implied oil volatility has not been applied in the oil volatility-sovereign risk nexus. Sharma and Thuraissamy (2013) examine the effect of oil uncertainty on CDS spreads, but they rely on the rolling variance of oil returns as a proxy for oil price

uncertainty. We contribute to the related literature by using oil implied volatility data within the recently introduced approaches of White et al. (2015) and Han et al. (2016), which are explained in the following section.

3. Employed methodologies

This study applies White et al.'s (2015) VAR for VaR model to study the degree of tail dependence between BRICS sovereign CDS spreads and the OVX. This econometric technique appears particularly suitable for studying sovereign risk reactions to oil volatility shocks by examining the pseudo impulse-response functions (IRFs), which allow for distinguishing between the reaction of sovereign risk to upper and lower quantile events. It is important to note that the multivariate quantile regression of White et al. (2015), the so-called VAR for VaR, allows estimating the tail dependence among the random variables of interest directly rather than recovering it indirectly via models of time-varying first and second moments as in the case of GARCH based models. Additionally, the multivariate regression quantiles overcomes the need for strong distributional assumptions as it directly models the (stationary) series. Although copula functions can model the dependence structure and provide information on the probability that two variables jointly experience extreme upwards or downwards movements, the application of copula functions is usually carried out based on Inference Function for Margins (IFS). Notably, the fitting of the marginal model requires care because if the model is unable to explain the time series variations, the residual (standardized residual) used in the copula estimation may provide spurious results (under/over fitting parameters). Furthermore, the approach of White et al. (2015) allows for the construction of pseudo impulse-response functions for the quantiles of the CDS spreads in BRICS in order to examine how oil volatility shocks are absorbed.

To measure the predictability of BRICS sovereign risk through OVX, we use bivariate CQ of Han et al. (2016). This technique provides a sample estimate of the measure of lead/lag dependence in quantiles and a corresponding test about directional predictability in quantiles from one series to another. Importantly, the CQs of Han et al. (2016) allows for capturing the directional quantile predictability of BRICS CDS spreads based on oil implied volatility in different quantiles, which cannot be captured by copula.

Detailed description of the above-mentioned methods is available in the Appendix.

4. Empirical findings

4.1. Data description

The data set used in this study consists of the daily price of the CBOE implied volatility index of crude oil (OVX) and the 5-year CDS spreads in Brazil, Russia, India, China, South Africa (BRICS) countries. An implied volatility index, such as the OVX, reflects both historical volatility information and investors' expectation on future market conditions (Liu et al., 2013). Accordingly, the OVX represents a forward-looking measure of oil market uncertainty. The CDS spread for the 5-year maturity contract is used as it represents the most traded and liquid contract. Data were collected from Thomson Reuters DataStream for the period from July 01, 2009, to March 31, 2017, for a total of 2023 daily observations. The beginning of the period starts just after the end of the US recession, as defined by the National Bureau of Economic Research (NBER), in order to ensure that the global financial crisis does not overlap with our sample period.

Table 1 presents the summary statistics of BRICS sovereign CDS spreads and OVX changes. The mean value of each country's CDS spread is higher than the mean value of OVX. The OVX standard deviation reflects different behaviour in each CDS market; it is highest (lowest) in Brazil (China), and it is smaller than that of CDS spreads. Furthermore,

Table 1

Descriptive statistics and unit root properties.

	Mean	Std. dev.	Skewness	Kurtosis	ADF	ADF (Δ)	KPSS	KPSS (Δ)
OVX	34.645	10.949	0.507	3.269	−3.3781**	−29.20***	0.6134**	0.0412
Brazil	189.50	93.903	1.550	4.777	−1.413	−23.29***	3.5263***	0.0941
Russia	221.08	85.490	1.690	6.641	−2.628***	−29.45***	1.6280***	0.0786
India	220.39	71.324	0.748	2.983	−1.733	−27.36***	0.8688***	0.1152
China	92.877	23.411	0.752	3.310	−3.111**	−26.91***	0.9739***	0.0331
SA	188.89	53.159	1.081	4.107	−2.820*	−40.34***	3.5619***	0.0694

Note: Std. dev. stands for standard deviations. ADF and KPSS are empirical statistics of the Augmented Dickey and Fuller (1979) unit root tests and the Kwiatkowski et al. (1992) stationarity test, respectively. Δ indicates that the series are first differenced.

*** Denotes the rejection of the null hypotheses of unit root and stationarity at the 1% level.

** Denotes the rejection of the null hypotheses of unit root and stationarity at the 5% level.

* Denotes the rejection of the null hypotheses of unit root and stationarity at the 10% level.

the results from two unit root tests (Augmented Dickey-Fuller and Kwiatkowski, Phillips, Schmidt, and Shin) show that all series are stationary at first differenced and hence we used the first difference of OVX and CDS spreads for econometric analysis.

Fig. 1 displays the time trends in BRICS sovereign CDS spreads and OVX. The sovereign CDS spreads of Brazil, Russia and South Africa

exhibit a spectacular rise starting in mid-2014, when oil prices sharply decline, and peak during 2015–16. The OVX exhibits similar behaviour during this period. The sovereign CDS spreads of India and China experience a steady increase during the European debt crisis of 2010–13. This crisis also adversely affects Russia's sovereign risk given the strong economic ties between Russia and Europe.

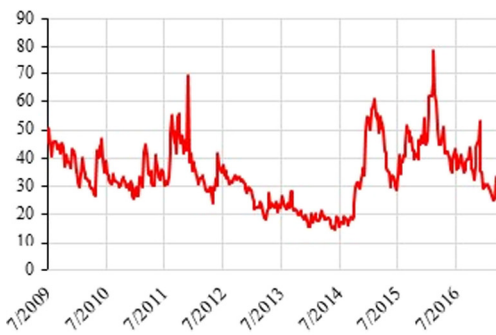
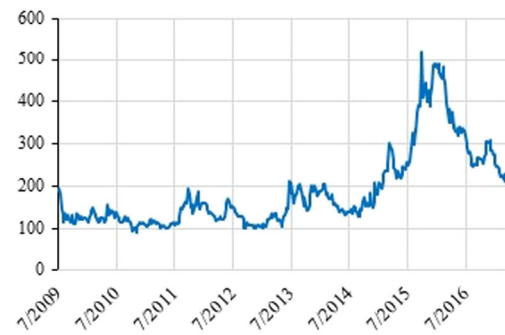
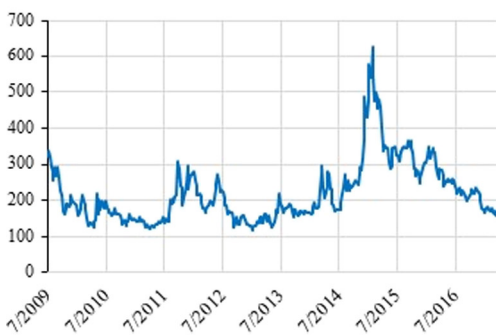
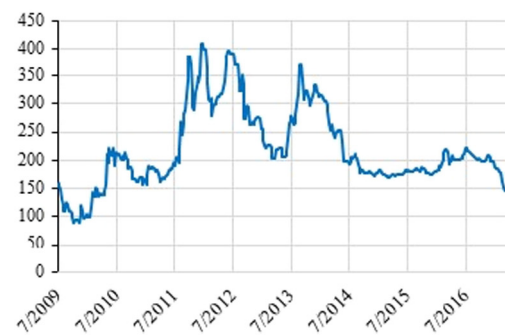
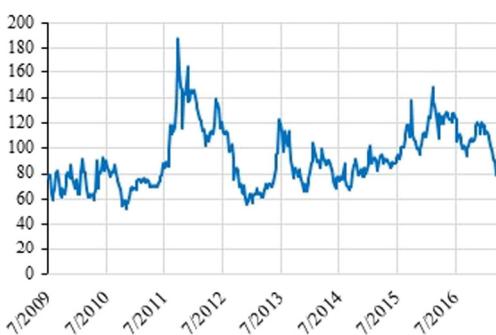
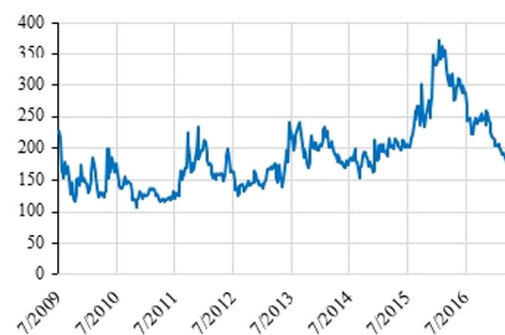
a). Oil volatility index (OVX)**b). Brazil****c). Russia****d). India****e). China****d). South Africa****Fig. 1.** Time trend of BRICS sovereign CDS spreads and oil volatility index.

Table 2
Estimates and standard errors for VAR for VaR.

Panel A: Downside spillovers ($q = 0.05$)					Panel B: Upside spillovers ($q = 0.95$)				
a). Brazil									
c_1 −0.216* (0.116)	a_{11} −0.236** (0.093)	a_{12} −0.164*** (0.052)	b_{11} 0.873*** (0.060)	b_{12} −0.118 (0.083)	c_1 0.146* (0.082)	a_{11} 0.167*** (0.047)	a_{12} 0.013 (0.091)	b_{11} 0.921*** (0.020)	b_{12} −0.053 (0.063)
c_2 −0.053** (0.023)	a_{21} 0.003 (0.010)	a_{22} −0.190*** (0.043)	b_{21} 0.085*** (0.007)	b_{22} 0.842*** (0.034)	c_2 0.129*** (0.050)	a_{21} 0.000 (0.008)	a_{22} 0.187*** (0.042)	b_{21} 0.041*** (0.004)	b_{22} 0.868*** (0.034)
b). Russia									
c_1 −0.469 (0.335)	a_{11} −0.492*** (0.162)	a_{12} −0.560** (0.334)	b_{11} 0.734*** (0.094)	b_{12} −0.181 (0.265)	c_1 0.993*** (0.352)	a_{11} 0.171*** (0.059)	a_{12} 0.801*** (0.169)	b_{11} 0.905*** (0.031)	b_{12} −0.627*** (0.182)
c_2 −0.064*** (0.020)	a_{21} −0.008* (0.004)	a_{22} −0.207*** (0.042)	b_{21} 0.084*** (0.005)	b_{22} 0.868*** (0.027)	c_2 0.162* (0.087)	a_{21} 0.014 (0.009)	a_{22} 0.166*** (0.057)	b_{21} 0.045*** (0.003)	b_{22} 0.876*** (0.050)
c). India									
c_1 0.004 (0.086)	a_{11} −0.058*** (0.016)	a_{12} −0.001 (0.078)	b_{11} 0.979*** (0.011)	b_{12} 0.000 (0.066)	c_1 0.107 (0.636)	a_{11} 0.183 (0.115)	a_{12} 0.235 (0.273)	b_{11} 0.923*** (0.088)	b_{12} −0.117 (0.258)
c_2 −0.073*** (0.020)	a_{21} −0.002 (0.004)	a_{22} −0.169*** (0.039)	b_{21} −0.005* (0.002)	b_{22} 0.887*** (0.024)	c_2 0.098* (0.061)	a_{21} 0.002 (0.010)	a_{22} 0.191*** (0.055)	b_{21} −0.006 (0.007)	b_{22} 0.887*** (0.036)
d). China									
c_1 −0.164* (0.087)	a_{11} −0.244*** (0.062)	a_{12} −0.118 (0.112)	b_{11} 0.856*** (0.047)	b_{12} −0.069 (0.073)	c_1 0.244* (0.133)	a_{11} 0.203 (0.179)	a_{12} 0.223*** (0.064)	b_{11} 0.874*** (0.102)	b_{12} −0.112* (0.070)
c_2 −0.065*** (0.025)	a_{21} −0.027 (0.026)	a_{22} −0.191*** (0.050)	b_{21} −0.015 (0.014)	b_{22} 0.875*** (0.033)	c_2 0.092 (0.080)	a_{21} −0.055* (0.030)	a_{22} 0.201*** (0.036)	b_{21} 0.048* (0.019)	b_{22} 0.828*** (0.046)
e). South Africa									
c_1 −0.446** (0.180)	a_{11} −0.277*** (0.067)	a_{12} −0.421*** (0.059)	b_{11} 0.802*** (0.053)	b_{12} −0.176** (0.085)	c_1 0.718 (0.545)	a_{11} 0.244*** (0.050)	a_{12} 0.365 (0.275)	b_{11} 0.773*** (0.049)	b_{12} −0.011 (0.357)
c_2 −0.040** (0.022)	a_{21} −0.002 (0.009)	a_{22} −0.189*** (0.039)	b_{21} 0.047*** (0.007)	b_{22} 0.859*** (0.029)	c_2 0.136* (0.083)	a_{21} −0.001 (0.015)	a_{22} 0.189*** (0.043)	b_{21} 0.055*** (0.019)	b_{22} 0.848*** (0.051)

Note: Estimated coefficients are in the first row. Standard errors are reported in italics in the second row. The coefficients correspond to the VAR for VaR model. ***, **, * denote significance at the 10%, 5%, and 1% levels, respectively.

4.2. Results from the VAR for VaR approach⁷

The estimates and standard errors for VAR for VaR are presented in Table 2, where the downside spillover (when $q = 0.05$) and the upside spillover (when $q = 0.95$) are displayed, respectively, in panels A and B. Each VAR structure relates the conditional quantile of oil volatility to the quantiles of CDS spread and vice versa. The upside and downside quantiles are explained by the predictors (parameters, e.g., a_{11} and a_{12}) as well as the pairs (a_{21} and a_{22}) of the first and second equations, respectively. To measure the tail dependence between oil volatility and BRICS sovereign CDS spreads, each quantile is expressed in terms of off-diagonal matrices b_{12} and b_{21} ; therefore, the null hypothesis of tail dependence is assessed by testing $b_{12} = b_{21} = 0$. The values of the diagonal autoregressive coefficients (b_{11} and b_{22}) are near unity and are statistically significant. This implies that VaR processes are autocorrelated, a finding that is in line with the reported literature that used the CAViAR models (see, e.g., White et al., 2015).

The b_{21} coefficient (in Panel A), which measures the impact of lower quantiles of oil volatility on the lower quantiles of sovereign risk, is positive and significant for Brazil, Russia, and South Africa. This indicates that an increase (decrease) in the expected future uncertainty in the oil market (the OVX is a forward-looking measure and thus represents the markets' consensus on the expected future uncertainty in the oil market) will increase (decrease) the sovereign risk in these three

countries during normal times.⁸ On the other hand, this b_{21} coefficient term is negative and significant for India and China, which implies that the expected future uncertainty in oil market will have an inverse (an increase in OVX will lead to a decrease in CDS) impact on sovereign risk. In cases of upside spillover (the results for upper quantiles as shown in Panel B), the b_{21} coefficient is positive and significant for Brazil, Russia, China and South Africa. An increase in oil uncertainty is expected to significantly increase the sovereign risk of these countries when both oil volatility and credit markets are in their upper quantiles. With the exception of India, the emerging markets' sovereign risk increases with the rising volatilities in the oil market, and the magnitude of this spillover is market-state dependent.

The results derived from the VAR for VaR regression are further enriched through a visual inspection of the impact of OVX quantiles on the lowest and highest quantiles of CDS spreads. The upside and downside standard VaR of CDS spreads and the VAR for VaR from OVX for both lower and upper quantiles are presented in Fig. 2. A careful investigation of the VaR and VAR for VaR plots reveals that the curves follow a homogenous pattern over the whole period. The upside/downside VaR curves of the CDS spread exhibit similar sequences of boom and busts in each market. For Brazil and Russia, CDS spreads show successive peaks during 2014–15, when oil prices plunge sharply. In contrast, India, China and South Africa show successive peaks during the European debt crisis of 2010–13. The VAR for VaR values are higher

⁷ The VAR for VaR approach of White et al. (2015) is implemented using the MATLAB codes available at <http://www.simonemanganeli.org/Simone/Research.html>.

⁸ We assume that oil volatilities and sovereign CDS spread will be lower during normal market states.

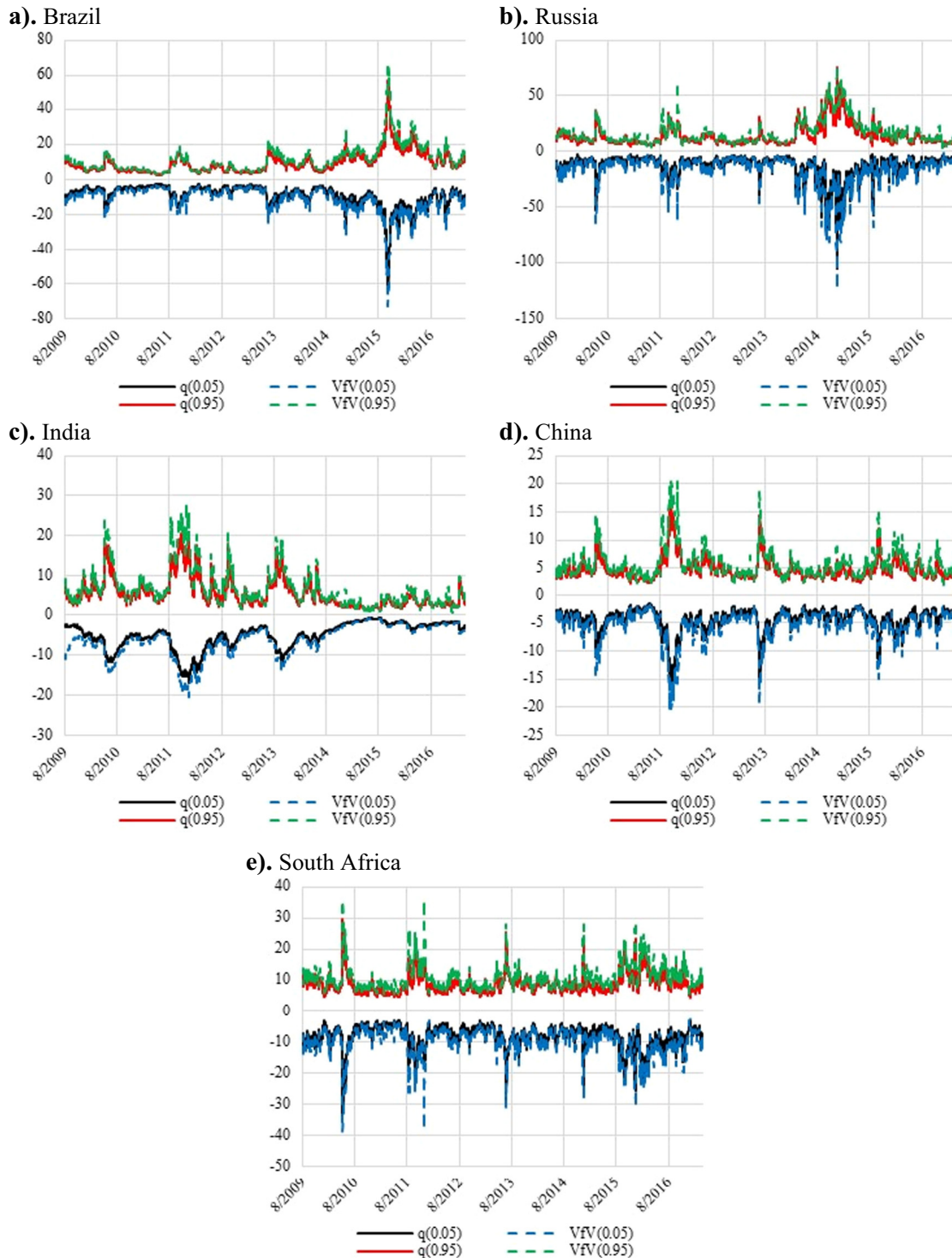


Fig. 2. Lowest and highest quantiles of CDS spread changes and impact of OVX quantiles. Note: These figures show 5% value at risk (VaR) of the BRICS sovereign CDS spread and 5% VaR for VaR from the oil volatility (OVX).

than the VaRs (for both lower and upper quantiles), and hence, a significant spillover from oil volatility to sovereign risk is evident.

4.3. Results from the pseudo QIRFs

Furthermore, we present in Fig. 3 the IRF of sovereign CDS spread changes to OVX using pseudo impulse response. Contrary to the standard IRF, which provides only the reaction of one market shock to another market, the pseudo IRF allows for distinguishing between the

reaction of a market to extreme (measured by upper quantiles) and tail (measured by lower quantiles) events. The lower and upper quantiles of oil volatility and sovereign risk show low and high uncertainty (risk), respectively. The horizontal axis reflects the time in days, while the vertical axis reflects the amplitude of the CDS spread's response to the OVX. The pseudo IRF tracks how the OVX negative/positive changes propagate through the BRICS sovereign CDS spread changes and how long it takes for BRICS sovereign CDS spread changes to completely absorb the OVX changes. Thus, the shock is completely

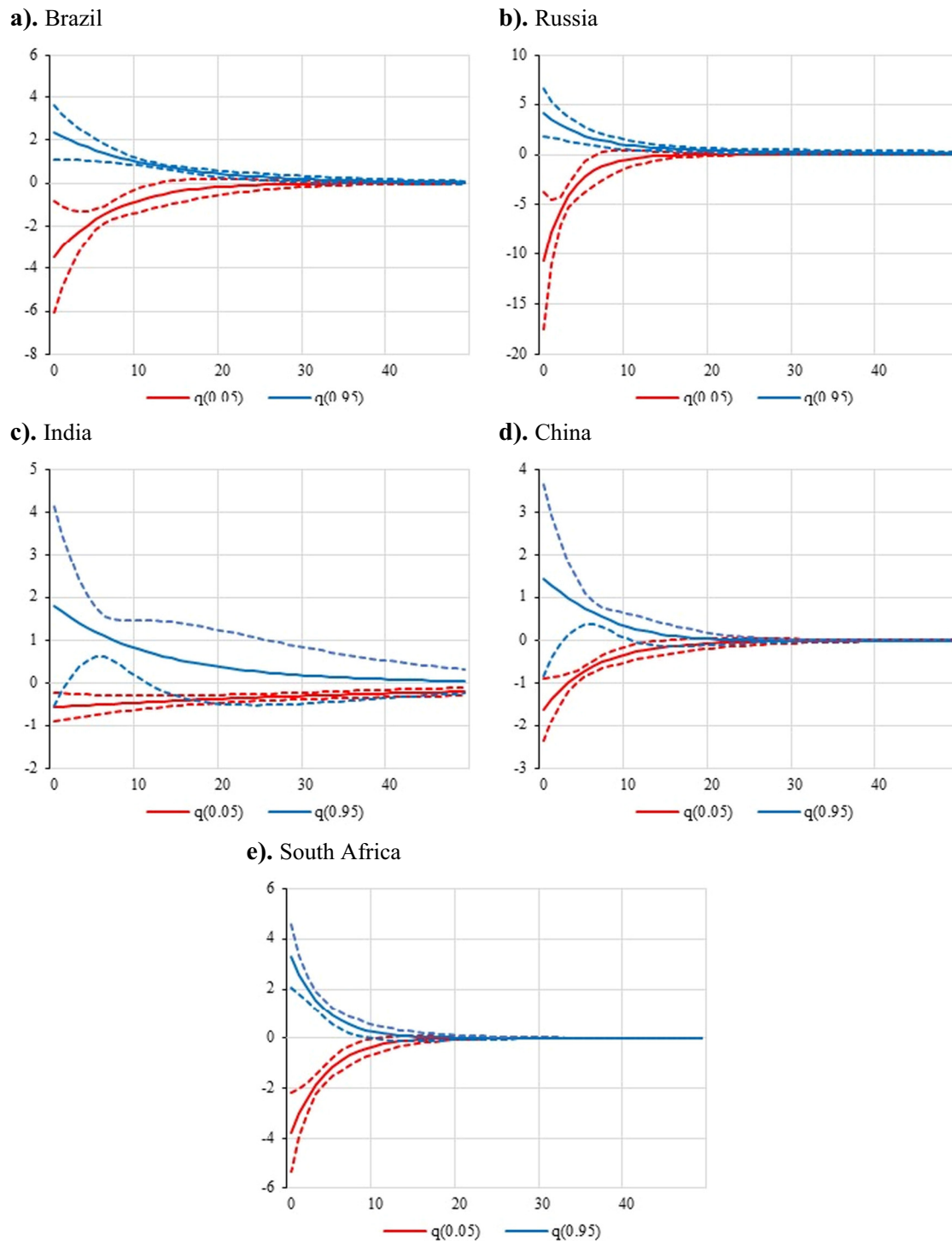


Fig. 3. Impulse-response functions of sovereign CDS spread change to oil volatility index. Note: The above figures show the pseudo quantile impulse-response functions for lowest (0.05) and upper (0.95) quantile-levels along with respective 95% confidence intervals. The magnitude of response is on y-axis while the period in days is on x-axis. These quantile impulse responses are drawn from a bivariate VAR for VaR and show how quantiles of SCDS spreads response to oil volatility shocks.

absorbed when the curves merge entirely with the horizontal axis, i.e., converge to zero.

The pseudo IRFs reveal numerous interesting findings. The sovereign CDS spreads of India and China react similarly to positive OVX shocks, with responses of 0 to 2. For Brazil, Russia and South Africa, the response to positive OVX shocks lies between 0 and 5. Furthermore, positive OVX shocks impact the BRICS sovereign CDS markets for approximately 18 days, and their effect is completely absorbed in 20 days as the IRF curves are aligned with the horizontal axis, in the cases of Brazil, Russia, China

and South Africa. However, for India, it takes about 40 days to completely absorb the effects of oil shocks. In sum, the findings from the pseudo IRFs show that the decrease in the sovereign risk of BRICS due to negative OVX shocks is more compared to the increase in sovereign risk resulting from positive oil shocks. Importantly, the findings are consistent with our findings reported above in the VAR for VaR section.

Table 3 reports the quantile values and the VAR for VaR of the sovereign CDS spread changes from OVX. The estimates are based on the bivariate VAR for VaR specification for the upside spillover and the

Table 3

Descriptive statistics of VaR of BRICS CDS and VAR for VaR from oil market.

Panel A): Downside spillovers			Panel B: Upside spillovers		
$q(0.05)$	$VfV(0.05)$	$H_0 = q(0.05) = VfV(0.05)$ $H_1 = q(0.05) > VfV(0.05)$	$q(0.95)$	$VfV(0.95)$	$H_0 = q(0.95) = VfV(0.95)$ $H_1 = q(0.95) < VfV(0.95)$
<i>a). Brazil</i>					
−9.172 (6.075)	−11.966 (7.502)	0.153 [0.000]	9.522 (6.219)	10.294 (7.091)	0.115 [0.000]
<i>b). Russia</i>					
−13.317 (10.481)	−17.183 (12.711)	0.175 [0.000]	13.994 (9.396)	16.308 (10.686)	0.156 [0.000]
<i>c). India</i>					
−5.085 (3.293)	−6.301 (3.987)	0.185 [0.000]	5.491 (3.790)	6.737 (4.919)	0.139 [0.000]
<i>d). China</i>					
−4.405 (2.162)	−5.330 (2.742)	0.210 [0.000]	4.803 (1.921)	5.847 (2.762)	0.212 [0.000]
<i>e). South Africa</i>					
−8.367 (3.803)	−10.902 (4.543)	0.179 [0.000]	8.938 (3.156)	9.903 (4.017)	0.143 [0.000]

Note: This table reports the quantile values and VAR for VaR of the sovereign CDS spread changes from the oil volatility index. Values in brackets are standard errors, and values in brackets are the p-values of the KS test.

downside spillover in Panels A and B, respectively. The estimated VAR for VaR (i.e., CoVaR) reflects the risk of BRICS sovereign CDS conditional on a significant shock from the oil volatility. In case of downside spillover ($q = 0.05$), the VaR and CoVaR values have negative signs. Notably, the CoVaR values are slightly higher than the VaR values, indicating that the downward movements in OVX significantly decrease BRICS sovereign risk. Similarly, in upside spillover (i.e., upper quantile, $q = 0.95$), the CoVaR values are higher than the VaR values. This implies that upward movements in OVX significantly increase the risk of BRICS sovereign CDS. Therefore, there is consistent evidence of a systematic upside risk spillover from oil volatility to BRICS sovereign risk. We infer that uncertainty in the oil market effectively propagates to the BRICS CDS markets. The differences across the upside/downside CoVaR conditional on OVX shocks may imply the presence of an intrinsic asymmetry in the mechanisms of shock transmissions.

4.4. Results from the bivariate CQ⁹

The significant directional predictability from OVX to BRICS sovereign risk is analyzed across a range of quantiles in this section. We use the bivariate CQ approach of Han et al. (2016), which offers flexibility in estimating the lead–lag relation between time series at different lags and quantiles simultaneously. The colour bar shows the magnitude of the Q-stats that test the significance of causal flows from OVX to CDS spreads. A careful look at Fig. 4 (lags = 5, 10 and 20) reveals some interesting findings. In Brazil, the extreme lower quantile (i.e., 0.05–0.20) of oil volatility predicts the lower quantile of CDS spreads, as indicated by the dark brown colours. This implies that low uncertainty in oil markets significantly Granger causes the lower risk of the CDS. No dependence is found at the median quantiles (0.40–0.60), where the colours are predominately green. However, the upper quantiles of oil volatility (0.80–0.95) predict the upper quantile of Brazil CDS spreads, indicating that high uncertainty in the OVX increases sovereign risk. In particular, we observe that the relationship between oil volatility and Brazil CDS spreads at the lower and upper quantiles is strengthened over the 10th and 20th lags. For Russia, the dark brown colour absolutely dominates over the whole range of quantiles, affirming the presence of a highly predictable Russian CDS through oil volatility at all lags. In India, the CDS spreads and OVX show some green coloured areas

located on the left side of their heat map plot, which represents the extreme lower quantile (i.e., 0–0.1) of oil volatility and the lower and upper quantiles of sovereign risk. This indicates that lower oil volatility cannot predict the low risk of CDS spreads, while the upper quantile of OVX significantly predicts the high risk of CDS spreads at all lags. For China, the dark brown colour is absolutely dominant at all lags and quantiles, except the extreme top left corner of the plot, which is green. This indicates that the low (high) volatility predicts low (high) CDS risk, while low oil volatility cannot predict the high risk of CDS. Finally, in South Africa, the resolution of the heat map plot of CDS spreads and OVX shows a similar colour pattern to that observed in Brazil and Russia. In fact, it shows that lower (upper) quantiles of OVX significantly predict the low (high) risk of CDS spreads. This main finding implies that downward (upward) movements in oil volatility significantly predict low (high) CDS spreads.

Moreover, the CQs of Brazil, Russia and South Africa show some green coloured areas located on the upper left and lower right corners that can be interpreted as a weak dependence between the oil and sovereign markets when the two markets are in different states (one market is bullish, while the other is bearish). Overall, the graphical results drawn from the cross-quantilogram approach highlight the ability of oil volatility to predict the sovereign risk of BRICS.

Distinguishing our study from prior studies via two novel quantile-based approaches – the multivariate quantile regression of White et al. (2015) and the directional quantile predictability of Han et al. (2016) – we reported more nuanced findings by uncovering the impact of oil implied volatility on BRICS CDS spreads in different quantiles. Specifically, we moved beyond the mean-based causality measures of Bouri et al. (2017a) and accounted for the difference in the impact of oil volatility on sovereign risk across lower, middle, and upper quantiles. Our findings nicely complement the findings from Sharma and Thuraiamy (2013) who overlooked the effect of oil uncertainty on BRICS sovereign risk – although BRICS include some of the largest oil exporters and oil importers – and focused only on six Asian countries (Indonesia, Japan, Malaysia, the Philippines, South Korea and Vietnam). We extended the work of Amstad et al. (2016) by focusing on the direct role of oil volatility, as a global risk factor, in explaining the sovereign CDS spread changes while differentiating between BRICs oil-importers and oil-exporters. Our findings also added to Lee et al. (2017) who considered only advanced economies and thus overlooked large emerging economies that are more dependent on energy commodities. While Liu et al. (2016) show that oil price volatility

⁹ The CQ of Han et al. (2016) is estimated using the R codes available at <http://www.oliverlinton.me.uk/research/software>.

can accentuate the volatility of country risk ratings, we revealed that downward and upward changes in oil volatility can predict, respectively, the low and high CDS spreads.

5. Conclusion

This study provided new insights to the oil volatility-sovereign risk nexus in BRICS countries and challenged the existing empirical findings on several grounds. First, we used implied volatility, which reflects oil

market uncertainty better than backward-looking volatility measures. Second, we uncovered for the first time the codependence between oil volatility and CDS spreads across the whole range of quantiles using the recently introduced approach of [White et al. \(2015\)](#), which allowed us to quantify the response of each BRICS country risk to negative/positive shocks in oil volatility through pseudo quantile IRFs. Third, we moved beyond the contemporaneous causality relationships and adopted a predictive approach in the Granger causality sense ([Han et al., 2016](#)) to more appropriately measure the directional quantile dependence between oil

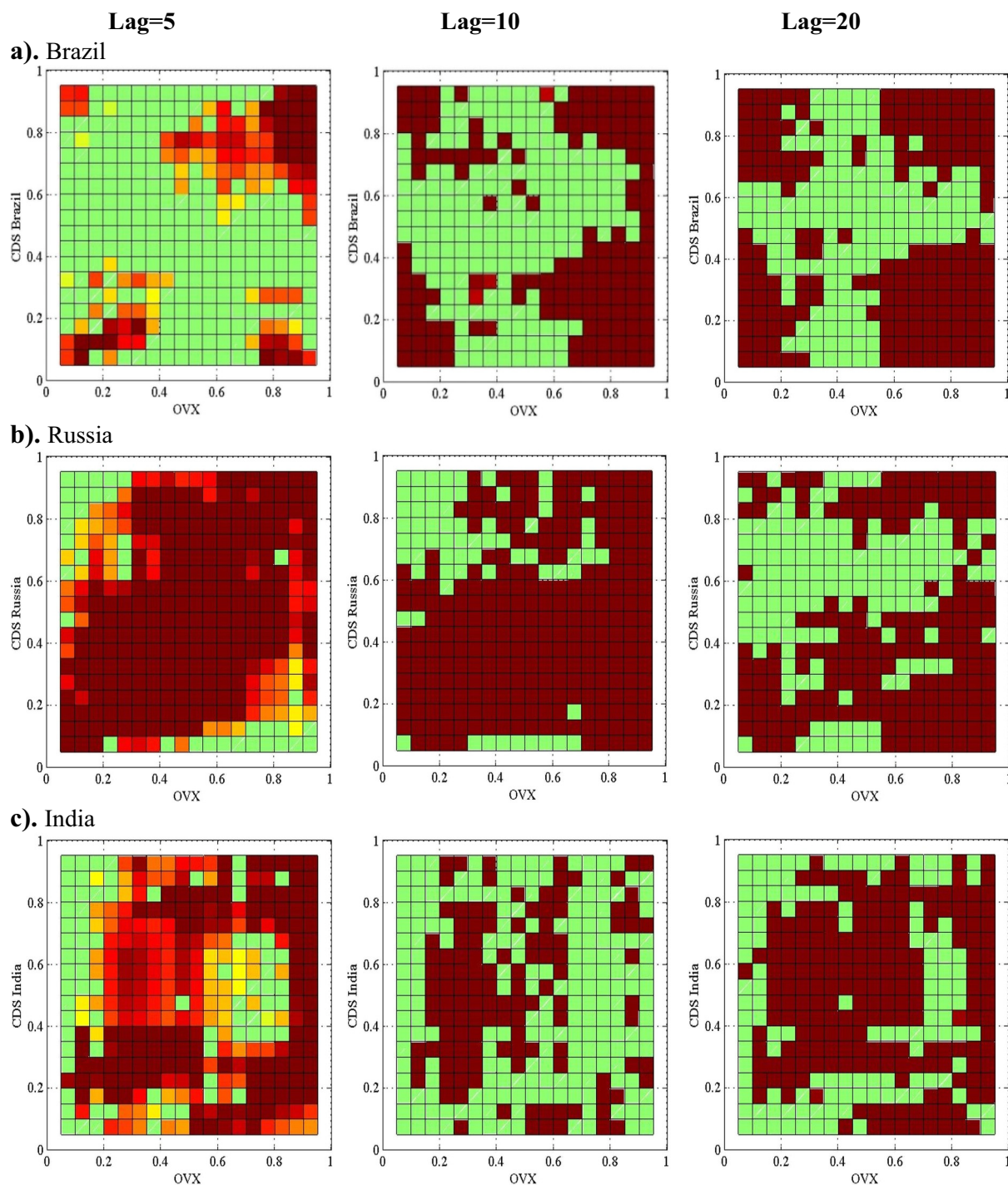


Fig. 4. Directional causality in quantiles from oil volatility (OVX) to BRICS sovereign CDS. Note: These figures show the significant directional predictability of BRICS sovereign CDS spread through oil volatility at various quantiles. X-axis corresponds to the quantiles of BRICS SCDS changes and Y-axis corresponds to quantiles of OVX. The colour bar, at the end of figures, shows the magnitude of the Q-stats testing the significance of causal flows from oil volatility to sovereign CDS.

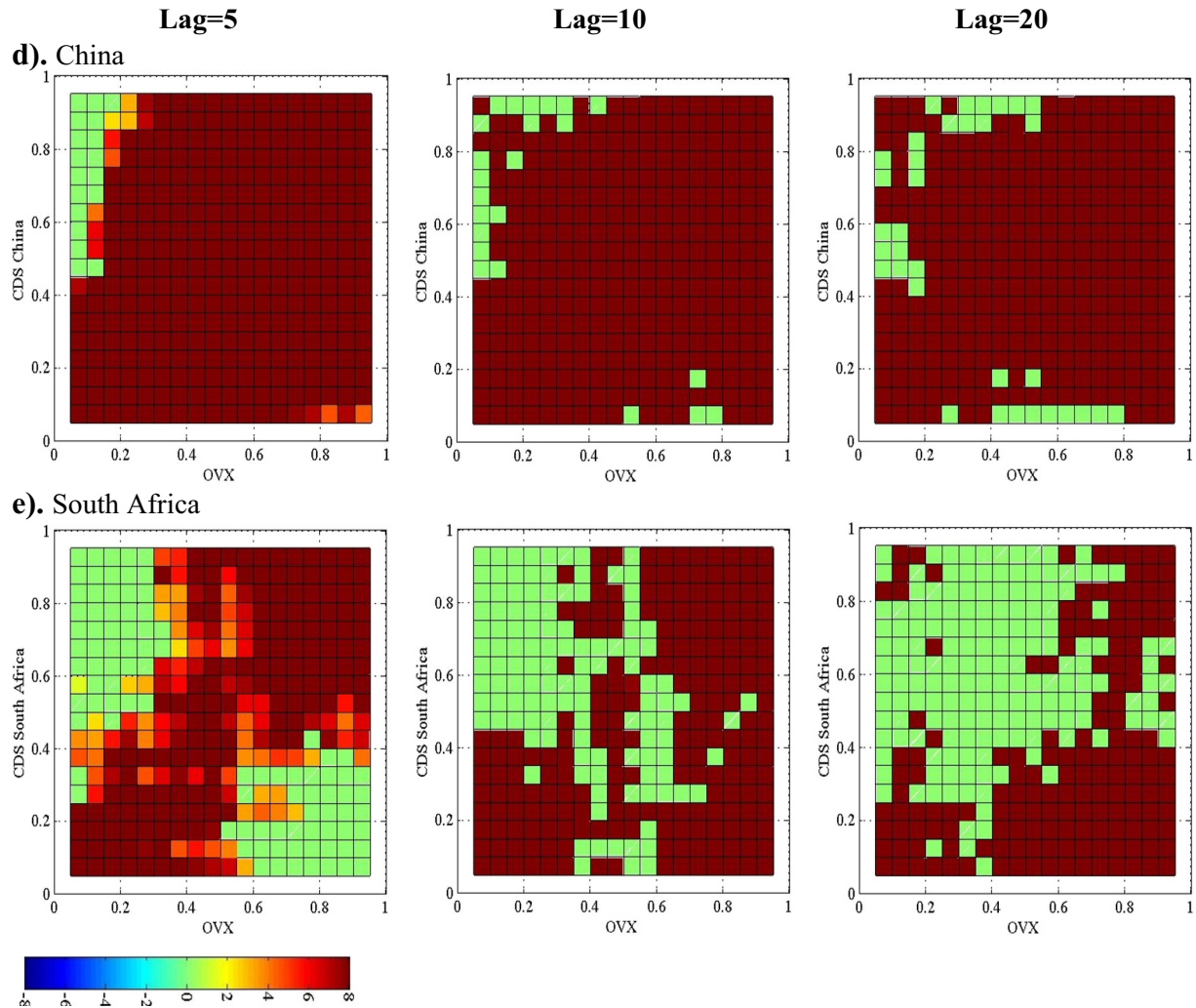


Fig. 4 (continued).

volatility and CDS spreads. Importantly, the cross-quantilogram approach of Han et al. (2016) is complementary to the VAR for VaR framework of White et al. (2015) since it gives detailed graphical assessments of the directional predictability from one market to another in various quantiles. The econometric methods employed enriched our understanding of which BRICS countries are more exposed to oil volatility shocks, especially during the oil market downturns of mid-2014, which entailed high levels of volatility. The empirical analyses revealed several noteworthy findings. First, we indicated the presence of a common component of systematic risk for both oil-exporting and oil-importing BRICS countries (Amstad et al., 2016), which is the volatility of crude oil. Second, we showed that oil exporters (Russia and Brazil) are more sensitive to positive shocks in oil volatility, whereas oil importers (China and India) are more sensitive to negative shocks. Third, we indicated that the mid-2014 oil price decline had intensified the reactions of BRICS sovereign risk to oil volatility in all BRICS countries. Fourth, we revealed the presence of an asymmetry in the mechanisms of shock transmissions. More specifically, the pseudo IRFs show that negative shocks in oil volatility decrease the CDS risk more than positive shocks increase CDS risk in the cases of Brazil, Russia and South Africa. Finally, the findings of directional predictability from oil volatility to BRICS sovereign risk revealed that the low (high) volatility of oil market predicts low (high) sovereign risk at various quantiles and lags.

The above findings are useful for policy-directed research and for traders/investors who can build upon our nuanced results to construct

hedging based strategy in relevant quantiles. For policy-makers, our results are useful for the design of suitable regulatory structures. Given the reported evidence on the increase in the volatility linkages in some cases (Brazil and Russia) following the considerable decrease in energy prices in mid-2014, it is crucial for policy-makers in those countries to devote more of their regulatory focus to the crude oil market in order to cushion adverse spillover effects. It is highly recommended that oil-dependent countries such as Russia diversify their economy away from energy and gas by promoting, among other things, tourism and tech-related and heavy machinery exportation. Furthermore, Russia can amass considerable foreign currency reserves that could help in running deficits for several years if necessary. Investors and traders who are exposed to severe economic losses from risk events in BRICS economies can more effectively re-weight their international portfolio if they better understand the effect of specific global risk factors (i.e., the volatility of crude oil) on the BRICS sovereign CDS spreads in the quantiles.

It is worth noting that although we documented significant effect of oil volatility on BRICS sovereign risk in the quantiles, we could not rule out the possibility that our findings might be affected by other economic, financial, and political factors. Importantly, our findings open the door for new in-depth research that explicitly accounts for the implied volatility of crude oil market in modeling sovereign CDS spreads.

Appendix A

This appendix details the different methods used in this study to examine the dependence between oil implied volatility shocks and BRICS sovereign risk.

A.1. The VAR for VaR approach

White et al.'s (2015) proposed the VAR for VaR regression quantiles approach to study the interdependence between the quantiles of random variables, which depend on the lagged quantiles, covariates and past innovations. This method has at least three advantages over standard techniques that rely on the parameterization of the entire multivariate distribution. First, quantile estimates obtained through regression are known to be robust to outliers, which is a desired feature for analyses of high-frequency financial data. Second, this regression quantile is a semi-parametric approach and as such imposes minimal distributional assumptions on the data-generating process (DGP). Third, it enables a direct examination of the tail dependence between random variables rather than recovering it indirectly through models of time-varying first and second moments. Thus, the effectiveness of the VAR for VaR approach is assessed while considering its setup for two random variables, i.e., X_{1t} and X_{2t} , on a given confidence level $\alpha \in (0, 1)$, where all information available at time t is given by F_{t-1} . Quantile q_{it} at time t of random variable X_{it} ; $i = 1, 2$, i.e., conditioned to F_{t-1} , can be written as

$$Pr[(X_{it} \leq q_{it} | F_{t-1}) = \alpha]; i = 1, 2 \quad (1)$$

The conditional quantiles of two random variables that follow a VAR specification can be written as

$$q_{1t} = X'_{it}\beta_1 + b_{11}q_{1t-1} + b_{12}q_{2t-1} \quad (2)$$

$$q_{2t} = X'_{it}\beta_2 + b_{21}q_{1t-1} + b_{22}q_{2t-1} \quad (3)$$

The VAR for VaR model of White et al. (2015) is formulated while considering the following assumptions:

The order of the quantile indices will be $0 < \alpha_{i1} < \dots < \alpha_{ip} < 1$ for $j = 1, \dots, p$, the α_{ij} th quantile of X_{it} conditional on F_{t-1} denoted $q^*_{i,j,t}$ is $q^*_{i,j,t} = \inf\{y: F_{it}(y) \geq \alpha_{ij}\}$, and if F_{it} is increasing, $q^*_{i,j,t} = F_{it}^{-1}(\alpha_{ij})$, $q^*_{i,j,t}$ is equal to

$$\int_{-\infty}^{q^*_{i,j,t}} dF_{it}(y) = E\left(1_{[Y_{it} \leq q^*_{i,j,t}]} \middle| F_{t-1}\right) = \alpha_{i,j}$$

where $dF_{it}(\cdot)$ is the Lebesgue-Stieltjes probability density function of X_{it} conditioned to F_{t-1} , and the stationary ergodic sequence of random $k \times 1$ vectors $[\psi_t]$ and real vectors $\beta^*_{ij} = (\beta^*_{i,j,1}, \dots, \beta^*_{i,j,k})$ and $\gamma^*_{i,j,t} = (\gamma^*_{i,j,t,1}, \dots, \gamma^*_{i,j,t,n})$; then, for each vector, $\gamma^*_{i,j,t,k}$ is $p \times 1$ vector such that for $i = 1 \dots n$ and $j = 1, \dots, p$, and all t

$$q^*_{i,j,t} = \psi_t' \beta^*_{ij} + \sum_{\tau=1}^m q^*_{t-\tau} \gamma^*_{i,j,\tau}$$

The multi-quantile version of the Conditional Autoregressive Value at Risk (MQ-CAViAR) of White et al. (2008) provides the basis for deriving the structure of the above equation. However, the bivariate data-generating process for the pair of variables, e.g., $Y_t = \{Y_{1t}, Y_{2t}\}$, is specified following White et al. (2015); it is written as.

$$\begin{bmatrix} Y_{1t} \\ Y_{2t} \end{bmatrix} = \begin{bmatrix} \alpha_t & 0 \\ \beta_t & \gamma_t \end{bmatrix} \begin{bmatrix} \epsilon_{1t} \\ \epsilon_{2t} \end{bmatrix}.$$

α_t , β_t and γ_t are F_t measurable, and each element of ϵ_t has the standard normal distribution and is mutually independent and

identically distributed (IID). The standard deviations of Y_{1t} , Y_{2t} are obtained by $\sigma_{1t} = \alpha_t$ and $\sigma_{2t} = \sqrt{\beta_t^2 + \gamma_t^2}$, respectively.

The mentioned α_t , β_t and γ_t follow the GARCH-type restrictions are as follows:

$$\sigma_{1t} = \tilde{c}_1 + \tilde{a}_{11}|Y_{1t-1}| + \tilde{a}_{12}|Y_{2t-1}| + \tilde{b}_{11}\sigma_{1t-1} + \tilde{b}_{12}\sigma_{2t-1}$$

$$\sigma_{2t} = \tilde{c}_2 + \tilde{a}_{21}|Y_{1t-1}| + \tilde{a}_{22}|Y_{2t-1}| + \tilde{b}_{21}\sigma_{1t-1} + \tilde{b}_{22}\sigma_{2t-1}$$

Furthermore, by substituting $\sigma_{1t} = \varphi(\alpha)q_{1t}$, from the above equation, the α th quantile that is related to the DGP takes the following form and is defined as the MQ-CAViAR (1,1):

$$q_{1t} = c_1(\alpha) + a_{11}(\alpha)|Y_{1t-1}| + a_{12}(\alpha)|Y_{2t-1}| + b_{11}(\alpha)q_{1t-1} + b_{12}q_{2t-1} \quad (4)$$

$$q_{2t} = c_2(\alpha) + a_{21}(\alpha)|Y_{1t-1}| + a_{22}(\alpha)|Y_{2t-1}| + b_{21}(\alpha)q_{1t-1} + b_{22}q_{2t-1} \quad (5)$$

Hence, the bivariate quantile is derived and can be written in a matrix form for the mentioned equations, $c_i(\alpha) = \tilde{c}_i \varphi^{-1}(\alpha)$, $a_{ij}(\alpha) = \tilde{a}_{ij} \varphi^{-1}(\alpha)$ and $b_{ij}(\alpha) = \tilde{b}_{ij}$, as

$$q_t = c + A|Y_{t-1}| + Bq_{t-1} \quad (6)$$

Parameters q_t , Y_{t-1} and c are bidimensional vectors, while A and B are the matrices whose components are shown in previous equation.

A.2. The pseudo QIRF and CoVaR

The one-off impulse function that is derived from the conditional quantile model is termed the pseudo IRF (White et al., 2015), which allows for capturing the impact of a one-off intervention at time t on a particular quantile. The given pseudo QIRF for Y_{it} is shown by $\Delta_{i,s}(\tilde{Y}_{1t})$ and can be written as

$$\Delta_{i,s}(\tilde{Y}_{1t}) = \tilde{q}_{i,t+s} - q_{i,t+s}, s = 1, 2, 3, \dots \quad (7)$$

where $\tilde{q}_{i,t+s}$ is the α th conditional quantile of the affected series. However, $\tilde{Y}_{i,t+s}$ and $q_{i,t+s}$ represent the α th conditional quantile of the unaffected series, i.e., $Y_{i,t+s}$. Specifically, White et al. (2015) defined the QIRF as follows:

$$\Delta_s(\tilde{Y}_{1t}) = \begin{bmatrix} \Delta_{1,s}(\tilde{Y}_{1t}) \\ \Delta_{2,s}(\tilde{Y}_{1t}) \end{bmatrix} \text{ and } D_t = |\tilde{Y}_t| - |Y_t| \quad (8)$$

The generalization of the QIRF is as follows:

$$\Delta_s(\tilde{Y}_{1t}) = AD_t \text{ for } s = 1 \text{ and } \Delta_s(\tilde{Y}_{1t}) = B^{(s-1)}AD_t \text{ for } s > 1 \quad (9)$$

We adopted White et al.'s (2015) VaR for VaR approach to measure CoVaR risk, where the CoVaR will be the VaR of the BRICS sovereign CDS conditioned to OVX decreases in its α th quantile; this is shown by q_i^α and can be written as

$$Pr(Y_j < \text{CoVaR}_{\alpha}^{(j|i)} | Y_i < q_i^\alpha) = \alpha \quad (10)$$

A.3. The bivariate CQ

Linton and Whang (2007) first proposed the univariate quantilegram to measure the directional predictability of stationary

time series at different quantiles. This univariate quantilegram was extended to the bivariate CQ by Han et al. (2016). The advanced CQ technique offers more flexibility in estimating the lead–lag relation between time series at different lags and quantiles simultaneously.

The two stationary time series can be defined as $\{x_{i,t}, t \in Z\}$, $i = 1, 2$. In the present study, $x_{1,t}$ and $x_{2,t}$ represent the BRICS sovereign CDS and oil volatility (OVX), respectively. However, the density functions and distribution of series $x_{i,t}$ are shown by $f_i(\cdot)$ and $F_i(\cdot)$. The quantile of $x_{i,t}$ is represented as $q_i(\alpha_i) = \inf\{v : F_i(v) \geq \alpha_i\}$ for $\alpha_i \in (0, 1)$, and the expression of two-dimensional series of quantiles are represented by $(q_1(\alpha_1)q_2(\alpha_2))^T$, for $\alpha \equiv (\alpha_1, \alpha_2)^T$.

The CQ for α quantile with k lags can be written as

$$\rho_\alpha(k) = \frac{E[\Psi_{\alpha 1}(x_{1,t} - q_1(\alpha_1))\Psi_{\alpha 2}(x_{2,t-k} - q_2(\alpha_2))]}{\sqrt{E[\Psi_{\alpha 1}^2(x_{1,t} - q_1(\alpha_1))]} \sqrt{E[\Psi_{\alpha 2}^2(x_{2,t-k} - q_2(\alpha_2))]}} \quad (11)$$

For $k = 0, \pm 1, \pm 2, \dots$ where $\Psi_a(\mu) \equiv 1[\mu < 0] - a$, $1(\cdot)$ denotes the indicator function, and $1[x_{i,t} \leq q_i(\alpha_i)]$ is called a *quantile hit*, or a quantile exceedance process. At different quantiles, the serial dependence between two time series is captured through Eq. (9). In the present framework, we measure the credit risk spillover to BRICS sovereign CDS from oil volatility and directional predictability through oil volatility by $\rho_\alpha(1)$ below or above a quantile $q_{ret}(\alpha_{ret})$ or $q_{vol}(\alpha_{vol})$ at time t . $\rho_\alpha(1) = 0$ indicates that the BRICS CDS risk is below or above quantile $q_{OVX}(\alpha_{OVX})$ at time t ; this, on average, will not provide useful information in predicting whether the CDS risk will be lower or higher than the quantile $q_{ret}(\alpha_{ret})$ on the next trading month. In contrast, $\rho_\alpha(1) \neq 0$ indicates a strong directional predictability generated through oil volatility to CDS at $\alpha = \alpha_{ret}(\alpha_{bs})$ or $\alpha_{vol}(\alpha_{OVX})$.

The CQ of the sample counterpart is given by

$$\hat{\rho}_\alpha(k) = \frac{\sum_{t=k+1}^T \Psi_{\alpha 1}(x_{1,t} - q_1(\alpha_1))\Psi_{\alpha 2}(x_{2,t-k} - \hat{q}_2(\alpha_2))}{\sqrt{\sum_{t=k+1}^T \Psi_{\alpha 1}^2(x_{1,t} - \hat{q}_1(\alpha_1))} \sqrt{\sum_{t=k+1}^T \Psi_{\alpha 2}^2(x_{2,t-k} - \hat{q}_2(\alpha_2))}} \quad (12)$$

In the above equation, $\hat{q}_i(\alpha_i)$ indicates the unconditional sample quantile of $x_{i,t}$, as proposed by Han et al. (2016). In addition to the quantile version of the Ljung-Box-Pierce statistic as $H_0 : \rho_\alpha(k) = 0$ for all $k \in 1, \dots, p$ against the alternative hypothesis of $H_1 : \rho_\alpha(k) \neq 0$ for $k \in 1, \dots, p$, under the following statistic:

$$\hat{Q}_\alpha^{(p)} = \frac{T(T+2) \sum_{k=1}^p \hat{\rho}_\alpha^2(k)}{T-k} \quad (13)$$

The asymptotic distribution of CQ contains “noise” under the null hypothesis of no directional predictability. Han et al. (2016) used the stationary bootstrap of Politis and Romano (1994) to approximate the null distribution and to conduct inferences.

This illustrates a portmanteau test statistic, $\hat{Q}_\alpha^{(p)}$, for directional predictability from one time series to another for up to p lags over quantile pair $\alpha = (\alpha_1, \alpha_2)$. In contrast to the usual bootstrap test, this test is a block bootstrap procedure that allows for handling inherent serial dependence in data by allowing random block lengths. Suppose that $B_{K_i, L_i} = \{(x_{1,t}, x_{2,t-k})\}_{t=k_i}^{L_i-1}$ is the i th block with length L_i starting from K_i ; then, L_i indicates an independent and identically distributed variable with $P_r(L_i = s) = \gamma(1 - \gamma)^{s-1}$, $s = 1, 2, \dots$ for $\gamma \in (0, 1)$.

Finally, in this framework, K_i is an IID sequence drawn from a uniform distribution $\{1, 2, \dots, T\}$. We replace the pair $(x_{1,t}, x_{2,t-k})$ by $(x_{1,t}, x_{2,t-k})$ with $j = k + (t \bmod (T - k))$ because the upper limit, B_{k_i, L_i} , may exceed the sample size, T , when $t > T$. Furthermore, the bootstrapped confidence intervals are constructed through pseudo re-sampling based on the

sequence of blocks and the associated portmanteau test statistic.

Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.eneco.2017.12.018>.

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